

Notes: Kim & Kung (RFS, 2017)

The Asset Redeployability Channel: How Uncertainty Affects Corporate Investment

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1 Big picture and incremental contribution

- **Question.** Why do some firms reduce investment much more than others after aggregate uncertainty shocks?
- **Core mechanism.** Uncertainty matters more when investment is more irreversible. Asset redeployability reduces irreversibility because more redeployable assets have higher liquidation values and can be sold or reused more easily in secondary markets.
- **Main empirical test.** After plausibly exogenous uncertainty shocks - Gulf War I and 9/11 - firms with less redeployable capital cut investment more than firms with more redeployable capital.
- **Main coefficient logic.** The key interaction is *Uncertainty Shock* $\times$ *Redeployability*. When uncertainty increases, the coefficient should be positive: high-redeployability firms reduce investment less, so their investment is higher relative to low-redeployability firms.
- **Long-panel result.** The same pattern appears in quarterly data from 1989 to 2009 using VIX and the economic policy uncertainty index as continuous uncertainty proxies.
- **Validation result.** More redeployable assets have higher recovery rates after default, are sold more actively in secondary asset markets, and support more leverage and shorter debt maturity.
- **Economic takeaway.** The investment-un uncertainty relation depends on real asset liquidity. Firms facing the same macro uncertainty shock respond differently because their capital can be unwound or resold with different ease.

1.1 Incremental contribution of the paper

- **New channel.** The paper identifies asset redeployability as a concrete real-side channel through which uncertainty affects investment. Earlier real-options work emphasized irreversibility; this paper measures a real-world source of irreversibility across firms and industries.
- **New measurement.** It constructs a BEA-capital-flow-based measure of redeployability by asking how widely an asset is used across industries and then aggregating those asset scores to the industry level using industry capital-expenditure weights.
- **New identification.** It combines exogenous aggregate uncertainty events with cross-sectional redeployability, rather than relying only on time-series correlations between uncertainty and aggregate investment.
- **New validation of the measure.** The paper does not simply assert that the redeployability index proxies for liquidation value. It validates the index using recovery rates, observed asset sales in secondary markets, leverage, and debt maturity.
- **New link across corporate finance literatures.** The paper bridges investment under uncertainty, asset liquidation values, capital structure, debt maturity, and used-asset-market liquidity.
- **Policy and empirical contribution.** It shows that the real effects of macro uncertainty are heterogeneous and can be predicted from the tradability/reusability of the firm's capital stock.

2 Data and measurement

2.1 Asset redeployability measure

The paper constructs asset redeployability using the 1997 Bureau of Economic Analysis capital flow table. This table decomposes capital expenditures of 123 industries into 180 asset categories. The idea is that an asset used by many industries has more potential buyers and therefore a more liquid secondary market.

For an asset category a , define an asset-level redeployability score as the fraction of industries or firms that use the asset. For industry j , aggregate asset scores using the industry’s capital expenditure weights:

$$Redeployability_j = \sum_{a=1}^{180} w_{ja} Redeployability_a.$$

- The baseline measure weights industries by market capitalization of public firms in that BEA industry.
- Alternative measures use equal weights, the number of firms, buyer financial constraints, and output-correlation adjustments.
- Manufacturing industries with more redeployable assets include leather, electrical equipment, and printing-related manufacturing; less redeployable industries include textiles, pulp and paper, plastics/rubber, and apparel.

Interpretation: The measure captures the size of the potential resale market for the capital a firm uses. It is a proxy for liquidation value and reversibility, not a direct measure of physical specificity alone.

2.2 Firm-level sample and investment outcome

The empirical investment sample uses Compustat quarterly firm observations. The main dependent variable is capital expenditure scaled by lagged total assets. Key firm controls include Tobin’s q , sales growth, cash flow, cash holdings, book leverage, age, tangibility, R&D, investment-grade rating, and financial-constraint indices.

Interpretation: The control variables separate the second-moment uncertainty channel from first-moment investment opportunities and financial constraints. Tobin’s q is especially important because it proxies expected investment opportunities, although measurement error in q is later addressed directly.

2.3 Uncertainty shocks and continuous uncertainty proxies

The paper uses two discrete uncertainty shocks:

- **Gulf War I:** event window November 1989-July 1991.
- **9/11:** event window December 2000-August 2002.

The two events are treated as plausibly exogenous to the investment opportunities of individual manufacturing firms.

The paper also studies a longer quarterly panel from 1989 to 2009 using continuous uncertainty proxies:

- the VIX index;
- the Baker-Bloom-Davis economic policy uncertainty index.

Interpretation: The event shocks give cleaner timing and exogeneity; the long panel checks whether the same mechanism operates more generally.

2.4 Validation outcomes for the redeployability measure

To show that the redeployability index is economically meaningful, the paper studies:

- ultimate recovery rates from Moody’s Ultimate Recovery Database;
- observed asset sales from Compustat and SDC Platinum;
- book leverage, market leverage, net leverage, and debt maturity.

Interpretation: These validation tests are important because the main investment result rests on interpreting redeployability as liquidation value. Higher recovery rates and more secondary-market asset sales support that interpretation.

3 Empirical specifications

3.1 Event-window difference-in-differences

The main event-window specification can be summarized as:

$$Investment_{it} = \alpha_i + \delta_t + \beta(After_t \times Redeployability_i) + \Gamma'X_{it} + \varepsilon_{it}.$$

- $Investment_{it}$ is capital expenditure divided by lagged assets.
- $After_t$ equals one for the post-shock quarters in the event window.
- $Redeployability_i$ is fixed at its pre-event value.
- Firm fixed effects absorb permanent firm investment levels.
- Year-month fixed effects absorb common time shocks.

Interpretation: A positive β means high-redeployability firms invest more relative to low-redeployability firms after uncertainty rises. Equivalently, low-redeployability firms cut investment more.

3.2 Low-redeployability dummy specification

Some regressions replace the continuous redeployability measure with a low-redeployability indicator:

$$Investment_{it} = \alpha_i + \delta_t + \beta(After_t \times LowRedeployability_i) + \Gamma'X_{it} + \varepsilon_{it}.$$

Interpretation: The expected sign on $After \times LowRedeployability$ is negative. This version makes the economic comparison transparent: low-redeployability firms reduce capital expenditure after uncertainty shocks.

3.3 Dynamic event specification

The paper replaces the single post-shock indicator with event-time indicators:

$$Investment_{it} = \alpha_i + \delta_t + \sum_k \beta_k(D_k \times Redeployability_i) + \Gamma'X_{it} + \varepsilon_{it}.$$

The dummies include periods before the shock and several post-shock horizons.

Interpretation: This specification tests whether the redeployability effect appears only after uncertainty rises and whether it persists over subsequent quarters.

3.4 Long-panel uncertainty specification

For the 1989-2009 quarterly sample, the paper estimates:

$$Investment_{it} = \alpha_i + \delta_t + \eta(Uncertainty_t \times Redeployability_i) + \theta Redeployability_i + \eta Uncertainty_t + \Gamma'X_{it} + \varepsilon_{it},$$

where $Uncertainty_t$ is either VIX or EPU.

Interpretation: This specification tests whether the asset redeployability channel is a general property of uncertainty-investment dynamics, rather than only a feature of Gulf War I and 9/11.

3.5 Measurement-error and validation specifications

The paper checks measurement error in Tobin’s q using OLS-IV first-difference estimators and high-order cumulant estimators. It also estimates validation regressions such as:

$$RecoveryRate_k = \alpha + \beta Redeployability_j + controls + FE + \varepsilon_k,$$

$$AssetSales_{jt} = \alpha + \beta Redeployability_{jt} + FE + \varepsilon_{jt},$$

$$CapitalStructure_{it} = \alpha + \beta Redeployability_{jt} + controls + FE + \varepsilon_{it}.$$

Interpretation: These tests check whether redeployability behaves like a liquidation-value measure outside the investment regression itself.

4 Identification strategy

4.1 Core identification logic

The paper combines two sources of variation:

1. plausibly exogenous increases in aggregate uncertainty;
2. predetermined cross-sectional differences in asset redeployability.

The identifying assumption is that, absent the uncertainty shock, high- and low-redeployability firms would not have experienced differential investment changes at the same event time after controlling for firm and time fixed effects and firm-level controls.

Interpretation: The design is a difference-in-differences with a continuous treatment intensity. It does not require uncertainty to be randomly assigned across firms; it requires redeployability to predict differential sensitivity to a common uncertainty shock for the reason posited by theory.

4.2 Why Gulf War I and 9/11 help identification

Both events generate sharp increases in uncertainty that are plausibly unrelated to individual firms’ investment opportunities. The paper shows that measures such as the VIX and dispersion of GDP forecasts rise sharply around the event dates.

Interpretation: These events reduce reverse causality concerns: a single firm’s investment plans do not cause the Gulf War or 9/11, and the shocks are common across firms.

4.3 Controlling for first-moment investment opportunities

A major alternative explanation is that low-redeployability firms face worse expected investment opportunities after uncertainty shocks. The paper addresses this by controlling for Tobin’s q , sales growth, cash flow, and fixed effects; by testing measurement-error corrections for q ; and by showing that the interaction remains significant.

Interpretation: The paper’s interpretation requires the effect to operate through investment irreversibility, not simply through lower expected demand. The control and measurement-error exercises are therefore central.

4.4 Robustness logic

Robustness tests include:

- bootstrapped placebo event windows;
- dynamic event-time regressions;
- alternative redeployability measures;
- long-panel VIX and EPU specifications;
- measurement-error corrections in Tobin’s q ;
- evidence on recovery rates, asset sales, leverage, and debt maturity.

Interpretation: The paper triangulates the channel. The main investment result is strongest when it is read together with direct evidence that redeployable assets have higher recovery values and more active resale markets.

5 Tables and figures

5.1 Figure 1 and Table 1: Uncertainty shocks and the redeployability measure

Read: Figure 1 shows large increases in GDP forecast dispersion and VIX around Gulf War I and 9/11. Table 1 defines and summarizes the redeployability measure for manufacturing industries.

Interpretation: The pairing is useful because Figure 1 documents the shock dimension and Table 1 documents the cross-sectional exposure dimension. The empirical design needs both.

Economic message: The paper’s contribution is strongest because it pairs plausibly exogenous macro uncertainty variation with a pre-existing real-asset characteristic.

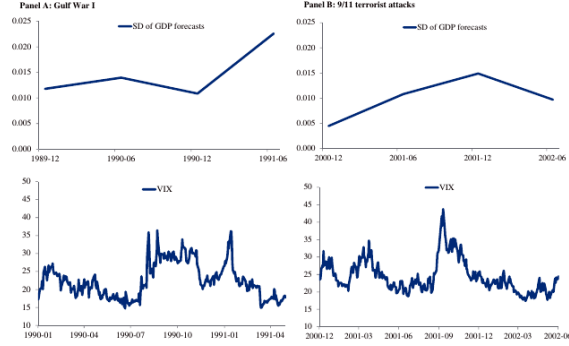


Figure 1
Changes in economic uncertainty around Gulf War I and 9/11 terrorist attacks
This figure plots the standard deviation of the nominal GDP forecast for one year ahead scaled by its mean from the Philadelphia Federal Reserve Bank's biannual Livingstone survey, and daily series of the VIX index for implied volatility of the S&P 500 index from the Chicago Board Options Exchange. Panel A plots these series for periods around Gulf War I, for which we define the event date as August 2, 1990, when Iraq invaded Kuwait, and Panel B plots for the terrorist attacks on September 11, 2001.

(a) Figure 1: uncertainty around Gulf War I and 9/11.

Table 1
Measure of redeployability of assets for manufacturing industries

Panel A: Descriptive statistics

	<i>N</i> industries	Mean	SD	Min	Q1	Median	Q3	Max
Asset redeployability	53	0.392	0.058	0.183	0.370	0.409	0.427	0.521

Panel B: Fifteen most and fifteen least redeployable industries among manufacturing

Most redeployable			Least redeployable		
Redeployability	Industry	Redeployability	Redeployability	Industry	
0.521	Leather & allied product manuf.	0.183		Textile mills	
0.461	Electric lighting equipment manuf.	0.240		Semiconductor & electronic component manuf.	
0.451	Printing & related support activities	0.274		Plastics & rubber products manuf.	
0.443	Soap, cleaning compound & toiletry manuf.	0.294		Pulp, paper, & paperboard mills	
0.443	Boiler, tank & shipping container manuf.	0.309		Textile product mills	
0.442	Agricultural chemical manuf.	0.316		Apparel manuf.	
0.442	Tobacco manuf.	0.331		Wood product manuf.	
0.439	Cutlery & handtool manuf.	0.341		Aerospace product & parts manuf.	
0.437	Paint, coating & adhesive manuf.	0.345		Computer & peripheral equipment manuf.	
0.433	Other chemical product & preparation manuf.	0.345		Electronic instrument manuf.	
0.433	Basic chemical manuf.	0.351		Furniture & related product manuf.	
0.432	Household appliance manuf.	0.353		Converted paper product manuf.	
0.430	Ordinance & accessories manuf.	0.359		Foundries	
0.427	Pharmaceutical & medicine manuf.	0.370		Iron and steel mills & manuf. from purchased steel	
0.426	Other transportation equipment manuf.	0.375		Industrial machinery manuf.	

This table presents statistics on the industry-level measure of asset redeployability for 53 manufacturing industries constructed using the 1997 Bureau of Economic Analysis (BEA) capital flow table merged with market capitalization of firms from Compustat in 1997. We construct the measure of asset redeployability as follows. First, we define an asset's redeployability score as the sum of market capitalization of all public firms in industries that use the asset over the sum of market capitalization across all public firms in the 123 industries in the BEA table. Second, we value-weight the asset-level redeployability scores across the 180 assets in the BEA table to give us an industry-level redeployability measure. Panel A reports descriptive statistics on the industry-level measure and Panel B reports the fifteen most and the fifteen least redeployable industries among manufacturing based on the measure.

(b) Table 1: redeployability measure.

5.2 Table 2 and Table 3: Descriptive characteristics and event-window investment effects

Read: Table 2 shows that high- and low-redeployability firms differ in size, q , leverage, and financial constraints, so controls matter. Table 3 shows the key result: after uncertainty shocks, low-redeployability firms reduce investment more, while the interaction $After \times Redeployability$ is positive and significant.

Interpretation: Table 3 is the main causal evidence. The coefficient on $After \times Redeployability$ is 0.876 in Gulf War I and 2.465 in the 9/11 sample before controls, and remains positive with controls. The low-redeployability dummy version gives the same conclusion.

Economic message: Investment irreversibility matters for how firms respond to uncertainty. Firms whose assets are hard to redeploy wait more when uncertainty rises.

Table 2
Descriptive statistics on firm characteristics

Panel A: Gulf War I					
Variable	Full Sample		High Redep.	Low Redep.	High – Low
	Mean	Std. Dev.	Mean	Mean	Diff.
Log(Total assets)	4.344	2.143	4.462	4.227	0.234
Age	13.135	8.578	13.809	12.461	1.348
Tobin's q	1.641	1.207	1.774	1.509	0.265
Sales growth	0.048	0.331	0.057	0.039	0.018
ROA	0.011	0.048	0.013	0.010	0.003
Cash flow	0.011	0.047	0.011	0.010	0.001
Cash holding	0.120	0.161	0.117	0.123	-0.006
Book leverage	0.262	0.202	0.275	0.249	0.026
Capex	0.015	0.013	0.015	0.016	-0.001
Tangibility	0.276	0.161	0.286	0.266	0.020
R&D	0.013	0.027	0.011	0.016	-0.005
Investment grade rating	0.114	0.317	0.131	0.096	0.034
SA index	-2.575	1.019	-2.633	-2.516	-0.118
WW index	-0.216	0.122	-0.226	-0.205	-0.020
Redeployability	0.392	0.063	0.439	0.346	0.093***
Redeployability (EW)	0.326	0.053	0.364	0.289	0.075***
Redeployability (FC)	0.390	0.063	0.437	0.344	0.093***
Redeployability (R^2)	0.154	0.025	0.172	0.136	0.036***
Observations	1,991	–	995	996	–
Panel B: 9/11 Terrorist attacks					
Log(Total assets)	4.969	2.101	4.822	5.115	-0.293
Age	15.073	10.426	14.733	15.413	-0.680
Tobin's q	1.905	1.527	2.051	1.759	0.293
Sales growth	-0.029	0.408	0.011	-0.069	0.079*
ROA	-0.012	0.062	-0.016	-0.008	-0.008
Cash flow	-0.013	0.067	-0.017	-0.010	-0.007
Cash holding	0.192	0.233	0.211	0.173	0.038
Book leverage	0.226	0.215	0.227	0.225	0.002
Capex	0.011	0.012	0.011	0.012	-0.001
Tangibility	0.231	0.168	0.216	0.247	-0.031
R&D	0.021	0.041	0.026	0.017	0.010
Investment grade rating	0.109	0.311	0.104	0.113	-0.009
SA index	-2.989	0.851	-2.930	-3.048	0.117
WW index	-0.234	0.119	-0.223	-0.246	0.022
Redeployability	0.371	0.055	0.413	0.330	0.083***
Redeployability (EW)	0.325	0.054	0.361	0.288	0.073***
Redeployability (FC)	0.370	0.055	0.411	0.328	0.082***
Redeployability (R^2)	0.215	0.030	0.238	0.193	0.045***
Observations	2,550	–	1,275	1,275	–

This table presents descriptive statistics on firm characteristics measured in the last quarter before the two episodes of a plausibly exogenous increase in aggregate economic uncertainty: Gulf War I in 1990 (Panel A) and the terrorist attacks on September 11, 2001 (Panel B). In each panel, the columns "High Redep." and "Low Redep." split the sample at the median of the redeployability measure based on market capitalization and separately report means for firms above and below the median redeployability. The column "High – Low" shows differences between the two columns. *, **, and *** represent differences significant at the 10%, 5%, and 1% levels, respectively. *Log(Total assets)* is the natural log of total book assets; *Age* is the number of years since IPO (proxied by the first appearance in Compustat); *Tobin's q* is the ratio of the market value of assets to the book value of assets; *Sales growth* is computed as the first difference of the natural log of sales; *ROA* is operating income before depreciation divided by lagged book assets; *Cash flow* is income before extraordinary items plus depreciations and amortizations divided by lagged book assets; *Cash holding* is cash and short-term investments divided by book assets; *Book leverage* is total debt divided by the sum of total debt and book equity; *Capex* is capital expenditures divided by lagged book assets; *Tangibility* is net PPE divided by book assets; *R&D* is research and development expenses divided by lagged book assets; *Investment grade rating* is coded as one if the bond rating from Standard & Poor's is above or equal to BBB-, and zero otherwise; *SA index* and *WW index* are the [Hadlock and Pierce \(2010\)](#) Size-Age and [Whited and Wu \(2009\)](#) indices of financial constraints; *Redeployability* is the measure of asset redeployability based on market capitalization of firms in industries; *Redeployability (EW)* is the measure of asset redeployability based on equal weights of industries; *Redeployability (FC)* is the measure of asset redeployability based on market capitalization of firms in industries and incorporating financial conditions of potential buyers; *Redeployability (R^2)* is the measure of asset redeployability based on market capitalization of firms in industries and incorporating correlations of outputs among industry peers.

(a) Table 2: firm characteristics.

Table 3
The effects of asset redeployability and increases in economic uncertainty on corporate investment

Dependent variable	(1)	(2)	(3)	(4)
	Capital expenditure / Assets			
Panel A: Gulf War I, November 1989–July 1991				
After	-0.150*** (-6.86)	-0.095*** (-3.93)	-	-
After × Low	-	-0.109*** (-2.93)	-	-
After × Redeployability	-	-	0.876** (2.49)	0.890** (2.58)
Tobin's q	-	-	-	0.176*** (5.08)
Sales growth	-	-	-	0.117** (2.42)
Cash flow	-	-	-	0.003 (1.27)
R^2	0.588	0.588	0.596	0.600
Observations	13,659	13,659	13,659	13,659
Panel B: 9/11 Terrorist Attacks, December 2000–August 2002				
After	-0.290*** (-4.97)	-0.202*** (-3.33)	-	-
After × Low	-	-0.174* (-1.79)	-	-
After × Redeployability	-	-	2.465** (2.63)	2.228*** (2.78)
Tobin's q	-	-	-	0.124*** (6.64)
Sales growth	-	-	-	0.151*** (5.06)
Cash flow	-	-	-	-0.002 (-0.52)
Firm fixed effects	Y	Y	Y	Y
Year-month fixed effects	N	N	Y	Y
R^2	0.571	0.572	0.580	0.589
Observations	17,564	17,564	17,564	17,564

This table examines the effect of asset redeployability and increases in uncertainty on corporate investment for the sample periods of Gulf War I (Panel A) and the 9/11 terrorist attacks (Panel B). The dependent variable is capital expenditure scaled by lagged total assets, a measure of corporate investment. The dummy variable *After* is equal to one for quarters ending after an increase in aggregate economic uncertainty, and zero otherwise. The dummy variable *Low* is equal to one if a firm's asset redeployability is below the median, and zero otherwise. *Redeployability* is held fixed for a given event window and measured in the latest fiscal year ending before or at the beginning of each event period. Definitions of the other variables are in [Table 2](#). All standard errors are adjusted for sample clustering at the industry level, and *t*-statistics are in parentheses. *, **, and *** represent results significant at the 10%, 5%, and 1% levels, respectively. Panel C repeats the estimation using 100 randomly selected sample periods consisting of seven consecutive quarters between 1989 and 2009, and reports the empirical distribution of the coefficients on *After × Redeployability*.

(b) Table 3: uncertainty, redeployability, and investment.

5.3 Table 4 and Figure 2: Dynamic effects and post-Gulf-War uncertainty decline

Read: Table 4 shows that the differential investment response appears after uncertainty rises, not before. Figure 2 shows uncertainty gradually declining after Gulf War I.

Interpretation: Dynamic effects support the timing of the mechanism. The negative coefficient on *Decrease × Redeployability* after Gulf War I means that low-redeployability firms increase investment more once uncertainty falls.

Economic message: Redeployability affects both sides of the uncertainty cycle: low-redeployability firms cut more when uncertainty rises and rebound more when uncertainty falls.

Table 4
Dynamic effects of economic uncertainty shocks on corporate investment conditional on asset redeployability

Dependent variable	(1)	(2)
	Capital expenditure / Assets	
Sample	Gulf War I	9/11 terrorist attacks
Before (-2) × Redeployability	-0.280 (-0.60)	-0.089 (-0.23)
Before (-1) × Redeployability	0.111 (0.22)	1.120 (1.40)
After (0) × Redeployability	0.573* (1.69)	2.339** (2.60)
After (1) × Redeployability	1.082** (2.60)	2.830*** (2.96)
After (2+) × Redeployability	0.846* (1.75)	2.589** (2.20)
Tobin's q	0.176*** (5.09)	0.123*** (6.75)
Sales growth	0.116** (5.42)	0.147*** (5.26)
Cash flow	0.003 (1.27)	-0.002 (-0.54)
Firm fixed effects	Y	Y
Year-month fixed effects	Y	Y
R^2	0.600	0.589
Observations	13,659	17,564

This table examines the dynamic effects of aggregate uncertainty shocks on corporate investment conditional on asset redeployability. The dependent variable is capital expenditure scaled by lagged total assets, a measure of corporate investment. The dummy variables *Before* (-1) and *Before* (-2) are equal to one if a sharp increase in aggregate uncertainty occurs in the next quarter and two quarters, respectively, due to Gulf War I (column 1) and the 9/11 terrorist attacks (column 2); the dummy variables *After* (0), *After* (1), and *After* (2+) are equal to one if a sharp increase in aggregate uncertainty has occurred in the current quarter, one quarter, and at least two quarters ago, respectively. *Redeployability* is held fixed for a given event window and measured in the latest fiscal year ending before or at the beginning of each event period. Definitions of the other variables are in Table 2. All standard errors are adjusted for sample clustering at the industry level, and t -statistics are in parentheses. *, **, and *** represent results significant at the 10%, 5%, and 1% levels, respectively.

(a) Table 4: dynamic effects.

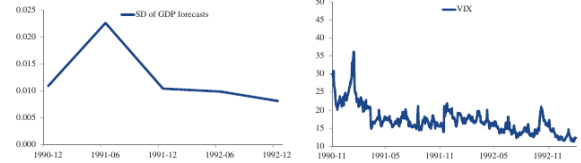


Figure 2

Changes in economic uncertainty during and after Gulf War I

This figure plots the standard deviation of the nominal GDP forecast for one year ahead scaled by its mean from the Philadelphia Federal Reserve Bank's biannual Livingstone survey, and daily series of the VIX index for implied volatility of the S&P 500 index from the Chicago Board Options Exchange for periods during and after Gulf War I. The war is generally considered to begin on August 2, 1990, when Iraq invaded Kuwait.

in uncertainty (1990.11 – 1991.07) as a period of high uncertainty, and the subsequent seven quarters (1991.08 – 1993.04) as a period of low uncertainty

(b) Figure 2: uncertainty during and after Gulf War I.

5.4 Table 5 Panels A–C: Long-panel VIX/EPU tests and decreasing uncertainty

Read: Panels A and B show that in the full 1989-2009 quarterly panel, interactions of uncertainty with redeployability are positive using VIX and generally positive using EPU. Panel B shows similar patterns for multi-segment firms. Panel C shows that when uncertainty declines after Gulf War I, low-redeployability firms increase investment more.

Interpretation: Table 5 expands the result beyond two event windows. The mechanism appears in continuous quarterly uncertainty variation, and it also works symmetrically when uncertainty falls.

Economic message: The asset-redeployability channel is not only a crisis-event pattern; it is a broader investment-un uncertainty relation.

Table 5
Asset redeployability, changes in aggregate uncertainty, and corporate investment

Panel A: Firm quarters from 1989 to 2009				
Dependent variable	(1)	(2)	(3)	(4)
Sample	Capital expenditure / Assets Firm quarters, 1989–2009			
VIX	−0.331*** (−3.89)	—	—	—
VIX × Redeployability	—	4.331*** (3.87)	—	—
EPU	—	—	−0.154*** (−5.20)	—
EPU × Redeployability	—	—	—	1.245** (2.02)
Redeployability	—	−1.554*** (−3.81)	—	−1.909*** (−2.85)
Tobin's q	0.185*** (9.60)	0.169*** (9.88)	0.182*** (9.89)	0.169*** (9.93)
Sales growth	0.241*** (7.01)	0.186*** (7.31)	0.239*** (7.23)	0.186*** (7.26)
Cash flow	0.012*** (4.20)	0.011*** (4.87)	0.012*** (4.28)	0.011*** (4.88)
Firm fixed effects	Y	Y	Y	Y
Year-month fixed effects	N	Y	N	Y
R^2	0.364	0.398	0.365	0.398
Observations	206,550	206,550	206,550	206,550
Panel B: Firm quarters from 1989 to 2009 for multi-segment firms				
Dependent variable	Capital expenditure / Assets			
Sample	Firm quarters, 1989–2009, multi-segment firms only			
Redeployability measure	Firm-specific	Industry-level	Firm-specific	Industry-level
VIX × Redeployability	5.359*** (3.85)	4.475*** (3.51)	—	—
EPU × Redeployability	—	—	0.803 (1.25)	0.657 (1.23)
Redeployability	−1.875*** (−3.16)	−1.565*** (−3.35)	−1.528* (−1.96)	−1.284* (−1.91)
Firm fixed effects	Y	Y	Y	Y
Year-month fixed effects	Y	Y	Y	Y
Time-varying controls	Y	Y	Y	Y
R^2	0.441	0.441	0.440	0.440
Observations	57,114	57,114	57,114	57,114

(continued)

Table 5
Continued

Panel C: During and after Gulf War I, November 1990–April 1993		(1)
Dependent variable	Capital expenditure / Assets	
Sample	During and after Gulf War I, November 1990–April 1993	
Decrease × Redeployability	—	−0.534* (−1.99)
Tobin's q	—	0.215*** (9.63)
Sales growth	—	0.184*** (3.95)
Cash flow	—	0.003 (0.76)
R^2	—	0.544
Observations	—	19,131

This table examines the effect of changes in aggregate uncertainty on corporate investment conditional on asset redeployability. The dependent variable is capital expenditure scaled by lagged total assets, a measure of corporate investment. Panels A and B use the full sample of firm quarters and a subsample of multi-segment firm quarters, respectively, from 1989 to 2009. Columns 1 and 2 of both panels use the VIX index as a proxy for aggregate economic uncertainty, and columns 3 and 4 use the economic policy uncertainty (EPU) index developed by Baker, Bloom, and Davis (2016) as a proxy for aggregate economic uncertainty. *VIX* and *EPU* are the average levels of the VIX and the EPU indices during a three-month period ending a month before a given quarter's ending date. In both panels, *Redeployability* is time-varying and measured in the latest fiscal year ending before a given quarter. In Panel B, columns 1 and 3 (2 and 4) use a firm-specific (industry-level) measure of asset redeployability. Panel C examines the effect of a decrease in aggregate uncertainty after Gulf War I on corporate investment conditional on asset redeployability. The dummy variable *Decrease* is equal to one for quarters ending after July 1991, and zero otherwise. In Panel C, *Redeployability* is held fixed for the event window and measured in the latest fiscal year ending before or at the beginning of the event period. Definitions of the other variables are in Table 2. Time-varying controls include Tobin's q , sales growth, and cash flow. All standard errors are adjusted for sample clustering at the industry level, and t -statistics are in parentheses. *, **, and *** represent results significant at the 10%, 5%, and 1% levels, respectively.

(a) Table 5 Panels A–B: full sample and multi-segment firms.

(b) Table 5 Panel C: decreasing uncertainty.

5.5 Table 6 and Table 7: Alternative redeployability measures and Tobin's q measurement error

Read: Table 6 shows that the main result survives several alternative redeployability measures. Table 7 uses OLS-IV and high-order cumulant estimators to address measurement error in Tobin's q .

Interpretation: Table 6 checks whether the result depends on a specific weighting scheme for asset redeployability. Table 7 checks whether the result is caused by poorly measured investment opportunities rather than uncertainty interacting with irreversibility.

Economic message: The effect survives both measurement of the key treatment variable and measurement of the most important control variable.

Table 6
Different measures of asset redeployability

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	Capital expenditure / Assets							
	M. cap & R ²				M. cap & R ²			
Redeployability measure	N. ind	N. firms	& Fin. con	& R ²	N. ind	N. firms	& Fin. con	& R ²
Panel A: Gulf War I and 9/11 terrorist attacks								
Sample	Gulf War I				9/11			
After × Redeployability	0.976** (2.22)	0.917** (2.38)	0.893** (2.58)	2.196** (2.40)	2.226*** (3.12)	2.220*** (3.08)	2.237*** (2.78)	3.792** (2.50)
R ²	0.600	0.600	0.600	0.600	0.589	0.589	0.589	0.589
Observations	13,659	13,659	13,659	13,659	17,564	17,564	17,564	17,564
Panel B: Firm quarters from 1989 to 2009								
VIX × Redeployability	4.752*** (4.74)	4.434*** (4.22)	4.351*** (3.87)	9.254*** (3.96)	—	—	—	—
EPU × Redeployability	—	—	—	—	1.141 (1.63)	1.124* (1.70)	1.254** (2.03)	2.172 (1.65)
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year-month fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Time-varying controls	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.398	0.398	0.398	0.398	0.397	0.398	0.398	0.398
Observations	206,550	206,550	206,550	206,476	206,550	206,550	206,550	206,476

This table examines robustness of the effect of asset redeployability and uncertainty on corporate investment for different measures of redeployability. The dependent variable is capital expenditure scaled by lagged total assets, a measure of corporate investment. Panel A (Panel B) reports estimates for Gulf War I and 9/11 (firm quarters from 1989 to 2009). Columns 1 and 5 employ equal weights for each industry to construct asset-level redeployability scores, and columns 2 and 6 employ the number of public firms in a given industry as the weight. Columns 3, 4, 7, and 8 employ the market capitalization of public firms in a given industry as the weight. In addition, columns 3 and 7 incorporate financial constraints of firms and columns 4 and 8 incorporate co-movement of outputs in a given industry to construct asset-level redeployability scores. Time-varying controls include Tobin's q , sales growth, and cash flow. Definitions of variables are as in Tables 2, 3, and 4. All standard errors are adjusted for sample clustering at the industry level and t -statistics are in parentheses. *, **, and *** represent results significant at the 10%, 5%, and 1% levels, respectively.

estimates in Table 6 Panel A (A.231 and A.245), and statistically significant for (a) Table 6: alternative redeployability measures.

Table 7
Continued

Panel B: High-order cumulant estimator

	(1)	(2)	(3)	(4)
Approach	EJW higher-order cumulant estimator			
Dependent Variable	Capital expenditure / Assets			
Sample	Gulf War I	9/11	1989-2009	1989-2009
After × Redeployability	0.883** (2.10)	2.005** (2.36)	—	—
VIX × Redeployability	—	—	1.324*** (3.32)	—
EPU × Redeployability	—	—	—	0.438* (1.87)
Redeployability	—	—	1.982*** (3.42)	1.572*** (3.22)
Tobin's q	0.556* (1.73)	0.767*** (4.97)	0.768*** (20.74)	0.768*** (20.74)
Sales growth	0.069 (1.02)	−0.086 (−1.13)	−0.043 (−1.00)	−0.050 (−1.22)
Cash flow	0.001 (0.23)	−0.005 (−1.42)	0.010*** (2.92)	0.010*** (2.91)
Firm fixed effects	Y	Y	Y	Y
Year-month fixed effects	Y	Y	Y	Y
ρ^2	0.077	0.168	0.151	0.153
τ^2	0.228	0.337	0.280	0.278
Observations	13,659	17,564	206,550	206,550

This table examines robustness of the effect of asset redeployability and uncertainty on corporate investment by addressing potential measurement error in Tobin's q using instrumental variables (IV) and high-order cumulant estimators. The dependent variable is capital expenditure scaled by lagged total assets, a measure of corporate investment. Panel A applies the OLS-IV estimator to first differenced equations using the third lags of q and cash flows as instruments for the first-differenced Tobin's q (Almeida, Campello, and Galvao 2010). At the bottom of the panel, we report estimates of the first differenced equation without using the instruments. Panel B uses the high-order cumulant estimator in Erickson, Jiang, and White (2014). ρ^2 captures how well true q explains investment, and τ^2 captures how well Tobin's q explains true q . Definitions of the other variables are as in Tables 2, 3, and 4. All standard errors are adjusted for sample clustering at the industry level, and t -statistics are

(a) Table 7 Panel B: high-order cumulant estimator.

Table 7
Measurement error in Tobin's q

Panel A: OLS-IV estimator

	(1)	(2)	(3)	(4)
Approach	OLS-IV on first-differenced equations			
Dependent variable	Capital expenditure / Assets			
Sample	Gulf War I	9/11	1989-2009	1989-2009
After × Redeployability	0.351 (0.99)	1.204*** (2.58)	—	—
VIX × Redeployability	—	—	1.591** (2.38)	—
EPU × Redeployability	—	—	—	0.359** (2.10)
Redeployability	—	—	0.318 (0.63)	0.318 (0.64)
Tobin's q	0.148 (0.84)	0.106* (1.86)	0.150*** (4.03)	0.150*** (4.03)
Sales growth	0.110*** (2.72)	0.069*** (2.70)	0.068*** (5.83)	0.068*** (5.81)
Cash flow	−0.001 (−0.12)	−0.003 (−1.25)	−0.002* (−1.93)	−0.002* (−1.92)
Year-month dummies	Y	Y	Y	Y
R ²	0.020	0.016	0.016	0.016
Observations	9,509	11,951	191,773	191,773
Estimates from FD equation without IV				
After × Redeployability	0.355	1.205**	—	—
VIX × Redeployability	—	—	1.643**	—
EPU × Redeployability	—	—	—	0.386***
Tobin's q	0.127***	0.090***	0.097***	0.097***

(continued)

existing empirical approaches: (i) instrumental variables (IV) and (ii) high-order cumulant estimators. In the first approach, we use sufficiently lagged values of Tobin's q and cash flows as instruments for the first-differenced (b) Table 7 Panel A: OLS-IV measurement-error test.

Table 8
Relationship between asset redeployability, recovery rates, trading in asset markets, and capital structure

Panel A: Ultimate recovery rates from default

	(1)	(2)	(3)
Dependent variable	Ultimate recovery rate (Moody's preferred method)		
Sample	Defaulted instruments, 1989-2009		
Redeployability	0.413** (2.61)	0.290** (2.16)	0.385** (2.28)
Log loan amount	−0.002 (−0.73)	−0.007** (−2.54)	−0.003 (−0.32)
Rating	—	—	−0.010 (−1.20)
Year fixed effects	Y	Y	Y
Seniority fixed effects	N	Y	Y
Collateral rank fixed effects	N	Y	Y
R ²	0.160	0.281	0.378
Observations	1,404	1,404	545

Panel B: Asset transactions in secondary markets

	(1)	(2)
Dependent variable	Asset sales / Net PP&E	Asset sales / Net PP&E
Asset sales data	Compustat	SDC Platinum
Redeployability	0.056* (1.69)	0.526*** (3.02)
Year fixed effects	Y	Y
R ²	0.034	0.064
Observations	1,707	1,911

(continued)

(b) Table 8 Panels A-B: recovery rates and asset sales.

5.6 Table 8 and Table C.1: Validation of redeployability and cross-industry spread

Read: Table 8 shows that redeployability predicts higher ultimate recovery rates, more secondary-market asset sales, higher leverage, and debt maturity outcomes. Table C.1 expands the redeployability ranking beyond manufacturing to all BEA industries.

Interpretation: These tables validate the economic content of the redeployability measure. If redeployability really measures liquidation value, it should show up in recovery rates, asset sales, and debt capacity; Table 8 confirms that pattern.

Economic message: More redeployable assets are not merely easier to describe as general-purpose assets; they are actually traded more, recovered at higher rates, and support different financing choices.

Table 8
Continued

Panel C: Capital structure outcomes

Dependent variable	(1)	(2)	(3)	(4)
	Book leverage	Market leverage	Net leverage	Debt maturity
Sample	Firm-years, 1989–2009			
Redeployability	0.324*** (2.99)	0.178* (1.97)	0.594*** (3.31)	0.037 (0.75)
log(Total assets)	0.022*** (6.20)	0.012*** (5.15)	0.024*** (4.53)	0.021*** (19.71)
Tobin's q	0.001 (0.46)	-0.027*** (-9.16)	-0.024*** (-8.79)	-0.003*** (-3.26)
ROA	0.142*** (3.92)	0.018 (0.77)	0.335*** (3.48)	0.004 (0.22)
Tangibility	0.283*** (4.73)	0.206*** (5.08)	0.748*** (7.19)	0.111*** (5.21)
Altman's Z-score	-0.029*** (-10.91)	-0.010*** (-5.60)	-0.026*** (-6.07)	0.003*** (3.13)
Year fixed effects	Y	Y	Y	Y
R^2	0.098	0.191	0.200	0.080
Observations	113,053	113,053	113,049	94,023

This table examines the relations between asset redeployability and recovery rates after default (Panel A), trading activities in secondary asset markets (Panel B), and firm financial leverage and debt maturity choices (Panel C) using the full set of industries available from the BEA table, excluding the financial and utilities industries. In the table, *Redeployability* is time-varying and measured in the previous year relative to the dependent variable. In Panel A, the dependent variable is *Ultimate recovery rate*, which is the recovery value that creditors actually receive at the resolution to default scaled by the original values of debt obtained from Moody's Ultimate Recovery Database (URD). Ultimate recovery rates are based on Moody's preferred method among settlement, liquidity, and trading prices methods. *Log loan amount* is the log of original loan amount; *Rating* is Moody's bond rating mapped to numerical values: Aaa = 1, Aa1 = 2, ..., C = 21. In Panel B, column 1 uses total sales of property, plants, and equipment (SPPE in Compustat) scaled by lagged total net PP&E (PPENT in Compustat) for a given BEA industry-year as the dependent variable, and column 2 uses the sum of used asset sales from the SDC Platinum M&A database scaled by lagged total net PP&E from Compustat for a given BEA industry-year as the dependent variable. In Panel C, column 1 (column 2) uses *Book leverage* (*Market leverage*), defined as total debt (long-term plus short-term debt) divided by the sum of book (market) value of equity and total debt as the dependent variable. Column 3 uses *Net leverage*, defined as total debt minus cash and short-term investments divided by the sum of book value of equity and total debt as the dependent variable. Column 4 uses *Debt maturity*, defined as the fraction of total debt that is due after one year as the dependent variable. *Altman's Z-score* is Altman's measure of bankruptcy probability defined as in [Mackie-Mason \(1990\)](#). Definitions of the other variables are as in [Table 2](#). All standard errors are adjusted for sample clustering at the industry level, and t -statistics are in parentheses.

(a) Table 8 Panel C: capital structure.

Table C.1
Fifteen most and fifteen least redeployable industries from all sectors

Most redeployable		Least redeployable	
Redeployability	Industry	Redeployability	Industry
0.711	Civic, social, professional & similar organizations	0.072	Accommodation
0.666	Social assistance	0.102	Oil & gas extraction
0.649	Legal services	0.139	Water transportation
0.630	Lessors of nonfinancial intangible assets	0.153	Hospitals
0.616	Personal & laundry services	0.183	Textile mills
0.604	Employment services	0.194	Educational services
0.585	Electronic, commercial & household goods repair	0.196	Agriculture & forestry support activities
0.584	Automotive equipment rental & leasing	0.204	Religious, grantmaking and giving & social advocacy org.
0.582	Advertising & related services	0.207	Rail transportation
0.575	Management of companies & enterprises	0.212	Air transportation
0.558	Travel arrangement & reservation services	0.233	Fishing, hunting & trapping
0.554	Newspaper, book & directory publishers	0.240	Semiconductor & electronic component manufacturing
0.553	Machinery and equipment rental & leasing	0.272	Coal mining
0.551	Wholesale trade	0.274	Plastics & rubber products manufacturing
0.545	New & maintenance and repair construction	0.289	Metal ores mining

This table presents the fifteen most and the fifteen least redeployable industries among all BEA industries, except for financial and utilities industries, based on the measure of asset redeployability constructed using the 1997 BEA capital flow table merged with market capitalization of firms from Compustat in 1997. We construct the measure of asset redeployability as follows. First, we define an asset's redeployability score as the sum of market capitalization of all public firms in industries that use the asset over the sum of market capitalization across all public firms in the 123 industries in the BEA table. Second, we value-weight the asset-level redeployability scores across the 180 assets in the BEA table to give us an industry-level redeployability measure.

(b) Table C.1: redeployability across all sectors.

6 Interpretive synthesis

- **Theory-to-data link.** Real-options theory predicts that uncertainty depresses investment more when capital is more irreversible. The paper operationalizes irreversibility with asset redeployability.
- **Why asset markets matter.** Liquid secondary markets reduce downside losses from bad states, so firms using redeployable assets have less reason to wait.
- **Why the event design matters.** Gulf War I and 9/11 generate common uncertainty shocks that are plausibly not caused by firm-level investment opportunities.
- **Why validation is essential.** Redeployability could otherwise be a black-box index. Recovery rates and asset sales show that the index measures something economically tied to liquidation values.
- **Corporate-finance implication.** Asset redeployability affects investment, liquidation values, and capital structure simultaneously. Investment and financing policies are connected through real asset liquidity.

7 Short summary

- The paper studies how asset redeployability changes the effect of uncertainty on investment.
- It constructs a new redeployability measure from BEA capital flow tables.
- Gulf War I and 9/11 provide plausibly exogenous aggregate uncertainty shocks.

- Firms with less redeployable assets cut investment more after uncertainty rises.
- The effect is dynamic: it appears after the shocks and reverses when uncertainty declines.
- Results hold in a long panel using VIX and EPU and under alternative redeployability measures.
- Measurement-error corrections for Tobin's q do not eliminate the result.
- Redeployability predicts recovery values, secondary-market asset sales, leverage, and debt maturity.
- The incremental contribution is to link investment under uncertainty to real asset liquidity and validate that link in both investment and financing data.

8 Conclusion

8.1 Core conclusion

Kim and Kung show that aggregate uncertainty reduces investment more for firms whose assets are less redeployable. This supports the real-options view that investment irreversibility magnifies the effect of uncertainty, but gives that theory a concrete empirical measure based on the usability and salability of real assets.

8.2 Economic intuition

When uncertainty rises, firms face more downside risk. If assets can be resold or redeployed easily, the firm can partially recover its investment in bad states. If assets are hard to redeploy, investment is more irreversible, so the firm delays capital spending. This is why the same uncertainty shock has different investment effects across firms.

8.3 Incremental contribution in one sentence

The paper's distinctive contribution is to measure asset redeployability from capital-flow data and show that this real asset characteristic explains both the investment response to uncertainty shocks and independent measures of liquidation values and financing capacity.

8.4 Final takeaway

Uncertainty shocks do not affect all firms equally. Real asset liquidity - the ability to redeploy capital in secondary markets - determines how much uncertainty turns into delayed investment.